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Information extraction from scanned invoice images using text analysis and layout features



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ABSTRACT

While storing invoice content as metadata to avoid paper document processing may be the future trend, almost all of daily issued invoices are still printed on paper or generated in digital formats such as PDFs. In this paper, we introduce the OCRMiner system for information extraction from scanned document images which is based on text analysis techniques in combination with layout features to extract indexing metadata of (semi-)structured documents. The system is designed to process the document in a similar way a human reader uses, i.e. to employ different layout and text attributes in a coordinated decision. The system consists of a set of interconnected modules that start with (possibly erroneous) character-based output from a standard OCR system and allow to apply different techniques and to expand the extracted knowledge at each step. Using an open source OCR, the system is able to recover the invoice data in 90% for English and in 88% for the Czech set.

1. Introduction

Automatic invoice processing systems gain significant interest of large companies who deal with enormous numbers of invoices each day, due to not only their legal value requiring them to be stored for years but also for economic reasons. Cristani et al. in [1] offer a comparison of €9 per manually processed invoice and €2 per automated processing of one invoice based on surveys in 2004 and 2003 respectively. A 2016 report by the Institute of Finance and Management [2] suggested that the average cost to process an invoice was $12.90.

Several challenges on recognition and extraction of key texts from scanned receipts and invoices have been organized recently, e.g. the Robust Reading Challenge on Scanned Receipt OCR and Information Extraction (SROIE) at ICDAR 2019 [3] or the Mobile-Captured Image Document Recognition for Vietnamese Receipts at RIVF2021 [4]. Still, annotated benchmark invoice datasets are not generally available due to confidential information, and the published papers do not offer detailed dataset descriptions and error analyses of the content. Moreover, although receipts and invoices have some common attributes, their analyses differ vastly due to complex graphical layouts and richer content of the required pieces of information used in invoices. In their experiments with recently published Kleister datasets [5] that include financial reports of charity organizations and non-disclosure agreements, Stanisławek et al. found that results using state-of-the-art models drop vastly with long, complex layouts in comparison to SROIE. In 2006, Lewis et al. [6] published the IIT Complex Document Information Processing Test Collection (IIT-CDIP) based on the Legacy Tobacco Documents Library, containing roughly 40 millions scanned pages for evaluation of document information processing tasks. In subsequent works, Harley et al. [7] took random samples of 16 categories in IIT-CDIP, each with 25,000 images, for document image classification evaluation. Nevertheless, none of those has benchmark results or gold standard data for the information extraction task.

In this article, we present the latest development of the OCRMiner project aiming at automated processing of (semi-)structured business documents, such as contracts and invoices, based solely on the analysis of OCR processing of the document pages. The current system significantly extends the original pipeline [8] for information extraction. We detail the improvements in each module, i.e. more precise page segmentation as well as Long short-term memory (LSTM) based OCR engine in text recognition, coping with OCR errors in keywords using an approximate string matching algorithm, or recognizing named entities using BERT based models. The presented approach boosted the results for the English invoice dataset and adapted well for Czech invoices.

The OCRMiner system represents the documents as a graph of hierarchical text blocks with automatic modular feature annotations based on keywords, text structures, named entity processing and location parser. With these features, a rule-based method is built to extract the desired information. The core idea behind OCRMiner lies in the human-like approach to the document content analysis. Different from other works overviewed in the next section, OCRMiner employs rich annotation from several text analysis modules in combination with text positional attributes. It does not only consider the absolute position

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of the text in the page presented by the bounding box, but also the relation with other groups of text. OCRMiner is also able to cope with OCR errors using a similarity search keyword matching algorithm. The system is unique in distinguishing both local (company name, address, ...) and global information such as buyer/seller identification. The higher-level modules of OCRMiner (invoice page classification, block type detection, ...) are language independent and the system is thus portable to new languages.

The next section brings a detailed overview of the latest results in invoice information extraction. Section 3 details the insides of OCRMiner and its modules. In Section 4, we offer a detailed evaluation of the system based on two datasets containing invoices in the English and Czech languages from tens of different suppliers.

2. Related works

Previous works aiming at automated processing of OCR invoices,[1] rely on predefined sets of layout of text data associated with a specific invoice format. One of the first systems [9] used a specific programming language for the description of the frame representation language for structured documents (FRESCO). For both text blocks and tables, keywords are located first, then the data items are extracted according to the programmed rules. The system did not deal with other items without a unique syntax. The recognition rate was between 40%–60%.

Adaptive rule-based systems that are based on layout information and text keywords usually divide the rules into the *vendor dependent* and *vendor independent* knowledge [10,11]. Cesarini et al. in [11] introduces a multi-class invoice analysis system (IAS) that interconnects the *layout structure* with the *logical structure* of the analysed document based on rules divided into the above mentioned two levels of knowledge. A comparable approach is presented in the German system named SmartFix [12,13]. In this system, the invoices are classified into known and unknown classes based on layout graph using case-base reasoning technique (similar to [14]). Those systems need a high amount of annotated graph samples to reach an acceptable result.

A template-based approach is proposed in [15,16]. In [15], users have to provide one entry of a table or a list in a document, then the system tries to find repeated structures which have the same or similar format. These candidates are evaluated by the overall match quality in term of perceptual coherence features such as alignment, height, width, separation and gaps. In [16], their objectives are towards small companies or single practitioners who cannot afford a large training set nor a customized system. The key idea is grouping documents generated by the same template, then generating extraction rules for each group based on at least a minimal number of similar training examples. The final result is a combination of results from different indexers for fields with a fixed value (e.g. the sender), a fixed position (e.g. the invoice number), a variant position and value (e.g. the total) etc. User feedback is allowed to add annotated documents as a new training example. For at least one training document per template, the system reaches 78% F1 score.

One of the main problems of processing invoices is the huge variety of invoice layout formats used in practice. Although a known set of metadata (i.e. the invoice number, date, the total, seller and buyer) is obligatory in most cases, their layout and positions are unconstrained. Therefore, many invoice analysis systems consider a particular layout as a class, usually connected to a particular vendor. If an invoice is assigned the correct class, then this will boost the extraction performance. However, the system must manage a large number of classes which may grow over time due to new suppliers or to layout updates. Trying to avoid classifying invoice into known and unknown classes which could lead to errors by misclassification, Aslan et al. [17] apply the part-based modelling (PBM) approach from computer vision to invoice processing based on deformable compositions of invoice parts candidates obtained from machine learning based classification techniques. The presented evaluation of invoice block detection ranges from 69% to 87% with the average of 71% accuracy.

Applying the results from the deep learning area, the CloudScan system [18] uses recurrent neural networks (specifically the bidirectional long short-term memory network [19]) to learn the item dependencies based on end-user provided feedback. The system reaches an F1 score of 84%–89% when trained on more than 300,000 invoices. A follow-up system, named InvoiceNet [20], introduced a specific deep network architecture denoted as Attend, Copy, Parse. The architecture does not use word-level labels for the end-to-end information extraction task, but it relies on experimentally designed set of dilated convolution blocks for each of the operations. The results improved the CloudScan results for four of the seven evaluated item types.

Riba at al. [21] propose a method using Graph Neural Networks (GNNs) for tabular layout detection based on the classification of graph node embeddings. The method does not rely on the document content although a commercial OCR tool is used for encoding classifying entity textual attributes into numeric, alphabetic, or symbol. The feature vector for each detected entity (node of the graph) includes its bounding box position and a histogram of textual attributes. An edge is created between two nodes if and only if a straight horizontal or vertical line exists between bounding boxes of the two nodes without crossing any other, excluding long edges covering more than a quarter of the page. The best accuracy of classifying 8 regions including address, table, or header lies between 82.7% and 84.5%.

Regarding the task of scanned invoice classification, in [22] the authors proposed a method to classify invoices into three categories: handwritten, machine-printed and receipts, for the best selection of OCR approach in latter processing. The features are extracted using a pre-trained Deep Convolution Neural Network (CNN) Alexnet. The experiments with 1,380 documents showed that the K-nearest neighbours method achieved the best result when compared to Random Forest (RF) and Naive Bayes (NB) with 98.4% total accuracy. In a similar task, Jadli and Hain [23] test different combinations of pre-trained CNN models (including VGGNet, Inception V3, DenseNet, and MobileNet-V2), dimensional reduction techniques (Principal Components Analysis, Linear Discriminant Analysis), and classification algorithms with a small dataset of 200 samples involving electronic invoices, hand written invoices, bank checks and receipts. The best result is 96.1% using full features extracted using VGG19 and Logistic Regression classifier. Similar techniques are used for document image source classification with the results ranging from 93% to 98% depending on the dataset granularity [24,25].

CUTIE (Convolution Universal Text Information Extractor) [26] applied convolution neural networks to texts converted to semantic embeddings in a grid-like structure. In this work, position features are taken into account, however, the exact positional relations are only implied. In [27,28], the authors proposed LayoutLM(v2) for pre-training text, layout, and image. The idea is inspired by Bidirectional Encoder Representations from Transformers (BERT) [29]. Although the image feature is used and the model works well on downstream tasks such as form understanding [3], receipt understanding [30], or document image classification [7], same as in CUTIE system, the potential relationship between text segments has not been taken into consideration. In addition, sufficient data and time are required to pre-train the model. A very similar idea on using text along with its position based on transformer architecture, particularly BERT, is presented in [31]. These systems need large datasets for pre-training models, and smaller labelled datasets for fine-tuning. Such datasets are not available by now for non-mainstream languages like the Czech language.

Experiments by Hamdi et al. [32] with invoice information extraction of document type and number, dates, amounts, and currency

[1] We denote as "OCR invoices" all kinds of invoice documents processed as an image by an Optical Character Recognition tool. This includes both scanned paper documents as well as digital born documents such as PDF.

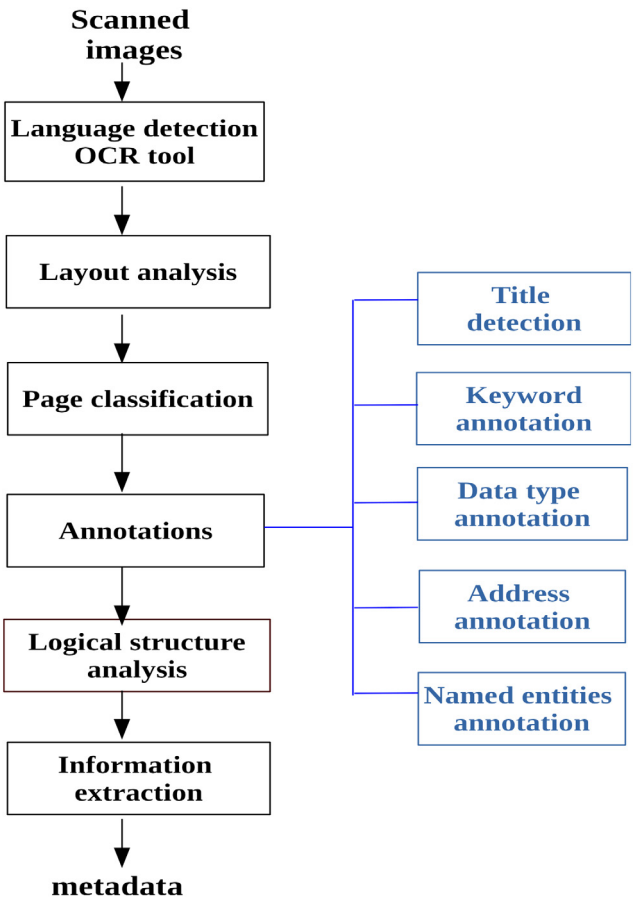


Fig. 1. The processing pipeline.

show that word classification approach (similar to [18]) outperforms sequence labelling methods [31] on most fields.

Recently, graph-based models have been proposed either using Graph Convolutional Networks [33,34] or Graph Attention Networks [35,36] for sequence labelling or node classification to identify the key items. The graph-based document representations include positional relations between text segments but they are limited to the direct neighbours only. Furthermore, training the graph embeddings needs a large amount of manually labelled document graphs.

A similar idea to the OCRMiner method that combines information from multiple sources of the data types and textual content in relative positions is found in [37]. In that system, for each field to be extracted, the generated candidates (e.g. dates, codes or numbers) are scored by their neighbourhood word and position embeddings. In contrast, OCRMiner begins with key phrases which drive the search for the data types in the visual surroundings inspired by the human-like reading of complex layouts. A disadvantage of the other method lies in limiting the neighbourhood to a fixed window. Since the layouts of invoices vary a lot, this approach can create unnecessary neighbours if the text is dense, and miss neighbours where the keyword and data are on the same line but distant. Moreover, OCRMiner not only deals with key texts matching fixed patterns, i.e. (invoice-, due-) dates or (invoice-, order-) numbers, but identifies also fields such as company names, contact persons, or general addresses.

3. The OCRMiner pipeline

A natural approach that humans use to extract specific information from a document involves 2 steps: scanning the document to locate the position of the information and then extract it in the desired format. Some types of information have unique structure defined by human rules, e.g. a VAT number. Other types need context to differentiate them from the same or other types of information, for instance, a date is distinctly identified as the invoice date or the due date by the keyword which goes with it. And others may need heuristic knowledge, for example the seller's name or address. The OCRMiner pipeline is generally inspired by this human approach.

An overall schema of the OCRMiner processing pipeline is illustrated in Fig. 1. The system is made of interconnected modules that allow to add any kind of partial annotations to the analysed document. First, the invoice image is processed by an OCR tool.[2] When OCRMiner needs to deal with multilingual documents, detecting languages used in the document is essential, not only for setting up OCR tool, but also for latter annotations. The main language of a document is defined as the language used by the title and data field names. For instance, in invoices, these are "invoice", "date", "order number", "payment method", etc. Therefore, given a dictionary of those terms for each possible language, if the document contains the terms of a specific language, then this language is assigned as the main language of the document. Otherwise, the language of the document is defined based on text language distribution detected by the OCR tool in the first run with a multiple languages setup.

Then, the physical document layout is built based on the word and character positional information from OCR (Layout analysis module). The Page classification[3] is performed with the information obtained by this step in the pipeline. From that point, a series of annotations using different natural language processing (NLP) techniques is applied to the content in the form of XML tagging, involving title, keywords, data types, addresses and named entities. These annotations provide semantic information and along with positional information, they create the context for further analysis. Based on these annotations, the logical structure, in which each text block is assigned an informational type, is extracted using a rule-based model. The final module extracts the desired metadata information.

3.1. Document physical layout analysis

The analysis of document structure can be divided into the *layout structure* (also called physical structure or geometric layout structure) and the *logical structure*. The layout structure analysis (or page segmentation) refers to the process of segmenting a document page into groups of text lines, text blocks, and/or graphical elements. In this paper, the physical layout structure is built using a bottom-up approach to obtain the document layout structure from the output of an OCR tool.

The invoice image is first processed by the OCR engine,[4] which computes the positions (bounding boxes) of each recognized word. Within the analysis, the words are grouped into lines based on three criteria: alignment, style, and distance. If two words have similar alignment and style, and the distance between them is less than a threshold then they are in the same line. The threshold was derived from the histogram of distances between each couple of adjacent words in 215 invoice images and currently corresponds to the height of the first word in the line.

The identified lines form (tabular) text blocks in a similar process, i.e. if two lines are vertically aligned, they have nearly the same font size, and the distance between them is less than a threshold, then they are grouped into a block. However, while the distances between words in a line usually correspond to the space character, the distances between lines in a block vary a lot depending on the graphical format. The best block-line threshold was set as twice as the height of previous line in the block.

[2] See [38] for details of the OCR setup.
[3] The OCRMiner page classification is described in detail in Section 3.2.
[4] The open source Tesseract-OCR [39] version 4.1.0 is used now.

Table 1
Evaluation of the invoice first page detection module.

Model	freq_words	title+ page#	freq_words+ title+page#	annotation+ title+page#	all features
Naive Bayes Classifier:					
Precision	93%	89%	95%	93%	95%
Recall	94%	96%	96%	**99%**	98%
F1 score	94%	92%	95%	96%	96%
Logistic Regression:					
Precision	97%	95%	**98%**	95%	97%
Recall	97%	96%	97%	97%	98%
F1 score	97%	95%	**98%**	96%	**98%**
Linear SVC:					
Precision	96%	96%	93%	90%	94%
Recall	96%	96%	87%	85%	92%
F1 score	96%	96%	88%	86%	93%

After deciding the document physical layout, each block receives an identification of its named positions in the form of block attributes, including the absolute position in the page and the relative position of the block to other blocks. The former divides a page into header, top, middle, bottom, and footer in the vertical direction, and left, right parts in the horizontal direction whereas the latter searches for the block's neighbours in the top, bottom, left, right and bottom-right positions respectively.

3.2. Page classification

The OCRMiner pipeline includes two classification tasks to answer the questions whether the document is actually an invoice or not and if the page is the first page. The reason is that business documents such as invoices are often scanned and processed in batches. OCRMiner thus contains the detection of an invoice first page. Similar to [40], the classification uses the available pieces of the semantic content of the document. The extent of the set of the available features depends on the module's position in the pipeline. Usually, this classification step goes between the layout analysis and the logical structure analysis. At this position, the features include binary features for the most frequent words obtained from the document collection for each language excluding entity names and stop words,[5] the (identified) page title (text, its position in the page and its size), and the (identified) page number. In case when the classification module goes further in the pipeline, the features are enriched with keywords and data type annotations as well as block types. While the *frequent words* feature considers only individual words, the annotations bring the logical meaning information. As an example, a part of the feature vector from an invoice first page looks as follows[6]:

```
{ u"dodavatel": True, u"odběratel": True,
  u"faktura": False, u"doklad": True,
  u"bankovní": True, u"splatnosti": True,
  u"zdanitelného": False, . . . ,
  "title": True, "top": 292, "height": 61,
  "page#": 1, "K_BUYER": True,
  "K_SELLER": True, "K_INVOICE NUMBER": True,
  "K_PAYMENT METHOD": True,
  "K_ORDER NUMBER": False, . . . ,
  "D_ORGANIZATION": True, "D_DATE": True,
  "general info": True, "buyer info": True,
  "seller info": True }
```

The classification output label is 1 if the page is a first page of an invoice, and 0 for others. The latter means the page can be second or further page of an invoice or not an invoice at all. The first page classification module was evaluated with a Czech dataset containing 1,150 pages (650 first invoice pages and 500 other pages[7]) using 10-fold cross validation. The experiments employed several fast classifiers, see Table 1 for the average precision, recall and F1 score of Support Vector Machines (Linear SVC), Logistic Regression, and Naive Bayes. The effect of the non-word features is clear in the group of SVC classifiers whereas the two other groups are not affected to such extent. Surprisingly, just the identified title and the page number can serve for 96% precision and recall. Moreover, by adding the annotations only (without the *frequent words*), the module can recognize up to 99% of the invoice first page. Among the classifiers, Naive Bayes reaches the highest recall of 99% while the Logistic Regression attains the best balance between the precision and the recall with the highest F1 score of **98%** (this is in accordance with the results of [23]). Other combinations of features and the visualization of the result are illustrated in Fig. 2.

3.3. The annotation modules

The annotation modules process the enriched OCR document and supplement its content with valuable pieces of information based on the output of natural language processing tasks. This part corresponds to the human reader procedure, who identifies e.g. a name of a city in various parts of the document and uses this information in the global-level rules.

The first three modules in the series of task-oriented annotators operate over plain text of recognized block lines to identify specific keywords and structured data such as dates, prices, VAT number, and IBAN. Keywords include words or phrases which are the signposts to find the desired piece of information. For instance, keywords for the invoice date are "invoice date", "date issued", "issued date", "billed on", or "date" at the beginning of a line. The set of keywords (per language) is obtained by analysing invoices in the development set. These modules can also cope with some OCR character-level errors.

The subsequent modules provide higher-level information such as named entities (personal names, organizations, locations) and formatted address specifications.

3.3.1. Named entity recognition

The invoice content analysis relies heavily on the ability to identify the relevant entities (typically person name, place name, organization, or a product name, artwork, date, time) mentioned in the tabular formatted texts. The task of named entity recognition (NER) consists of two steps: named entity identification (including named entity boundaries detection in case of multi-word NER) and the entity classification. OCRMiner currently employs pre-trained BERT NER model [41] based on the Google BERT-LARGE model [29] for English and the SLAVIC-BERT model [42] for the Czech language.

Although current state-of-the-art results in NER look promising for both English and Czech, the text style of invoices causes the general models to miss important context properties. First of all, the text chunks are rather short in invoice blocks when compared to the full sentences in existing models. The next challenge is that uppercase formatting (which is an important feature in the NER models for Czech and English) is used more frequently in this type of documents, for example, for headings or company names. The third problem is strong multilingual characteristics of many invoices, e.g. in English invoices, the names of organizations or streets can be in a different (local) language.

[5] In the experiment, these binary features (True/False meaning the word appears/does not appear in the text) are created for 111 words that resulted from choosing 150 most frequent words and eliminating entity names and stop words.

[6] The prefix u"" stands for a *frequent word*, "K_" for a *keyword* annotation, "D_" for a *data type* annotation. The presented frequent words are "dodavatel – seller", "odběratel – buyer", "faktura – invoice", "doklad – document", "splatnosti – due date", "zdanitelného – taxable".

[7] The dataset is a bit different from the one used in [38]. In the current experiment, pages which are not in Czech and extremely bad quality scans were removed and new data was added.

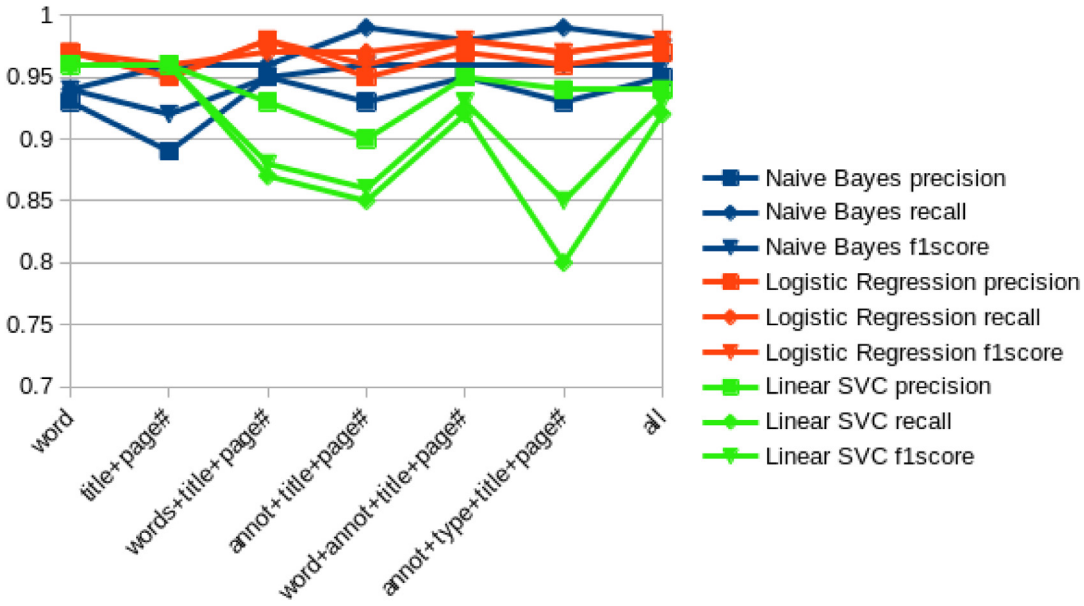


Fig. 2. Evaluation of invoice first page classifiers using different features.

3.3.2. Location names recognition

Beside the location entities detected by NER, OCRMiner uses a global address parser `libpostal` [43], a statistical model based on conditional random fields trained on Open Street Maps[8] and Open Addresses.[9] The reasons for this doubling reflect the importance of addresses in the invoice metadata and the need to be able to solve possible conflicts in the name classification. Each of the two modules has positive matches on different data. For example, street names such as *Mahenova 9/181* are well recognized by `libpostal` while the NER module recognizes *Mahenova* as person name. On the other hand, the location names recognition matches *Konica Minolta Business* as an office building in the U.S. while the NER module annotates it (correctly) as an organization name.

3.4. Block type detection

After the annotation modules add all relevant local pieces of information to the analysed content, each block is assigned possible block types. The invoice text blocks are categorized into:

- general information containing invoice number, order number, date;
- seller/buyer information including company name, address, vat number, contact detail;
- delivery information such as delivery address, date, method, code and cost;
- bank information covering the bank name, address, swift code, account number;
- the invoice title and page number.

Blocks that do not belong to any of these categories are assigned an empty label.

The OCRMiner block type detection technique is based on a set of logical rules that combine information obtained in the preceding pipeline steps. The rules are in human readable and easy to edit form (see an example in Fig. 3). Each rule is applied to each block in the invoice document. If a block meets the rule's condition, then the label is added to the block type. A detailed evaluation of this module with a prototype English dataset is provided in [8].

[8] http://openstreetmap.org/.
[9] http://openaddresses.io/.

3.5. Information extraction

The extracted data can be a single piece of information such as the invoice number, invoice date, the order number, the order date, the due date, the payment date, the payment method, IBAN, or the swift code, or a group of information such as the buyer information, seller information and the delivery address. The first group can be divided into smaller categories: "DATE", "PRICE", "NUMBER", and "GENERAL". The first two clusters usually attach keywords with data types whereas the two others often have no rule for the data. Unique structured information such as VAT number can be revealed without keywords.

Each type of information has a confidence score attached distinguishing the situations when only the keyword was found, the data type matches, or both.

For instance, with the VAT number as an example:

```
{"key": VAT NUMBER,
 "data": VAT NUMBER,
 "key_conf": 0.7,
 "data_conf": 0.8,
 "combine_conf": 1.0}
```

If both the keyword and the data type are found, then the confidence score is set to 1.0. Otherwise, if only the keyword or the data type is found for some reason (e.g. OCR errors), then the confidence is lowered to 0.7, or 0.8 respectively. The confidence score values are based on the development set evaluation.

The extraction workflow process is illustrated in Fig. 4. First, the system searches for the appropriate keywords in the annotations. This returns the block and the line containing the keyword if the keyword is found. Then, if the line does not contain the expected data, a weighted function is applied to the right block and bottom/bottom-right block to see if the data lies on the right of the keyword or under the keyword. This function sets a high score to the correct data type, a lower score to any other data found in the annotations, and a penalty for other keywords founds. The reason is that the information is often organized in columns where keywords are in the left column and data on its right. The higher weighted block is more likely to contain the data. If the keyword is not found, then the system will look at the data type in case it is structured information. The annotation steps are able to cope with some OCR errors, so, the validation step checks and corrects the

```
seller info -> block_annot.data in [ORGANIZATION, PERSON, LOCATION, CITY,
        COUNTRY, EMAIL, PHONE] and SELLER in top_blocks.block_annot.keyword
        and top_blocks.num_lines == 1
```

Fig. 3. Block type rule example: if annotations in the block contains more than at least one label in the set {ORGANIZATION, PERSON, LOCATION, CITY, COUNTRY, EMAIL, PHONE} and the top block has only one line marking the SELLER keyword, then the block type is `seller info`.

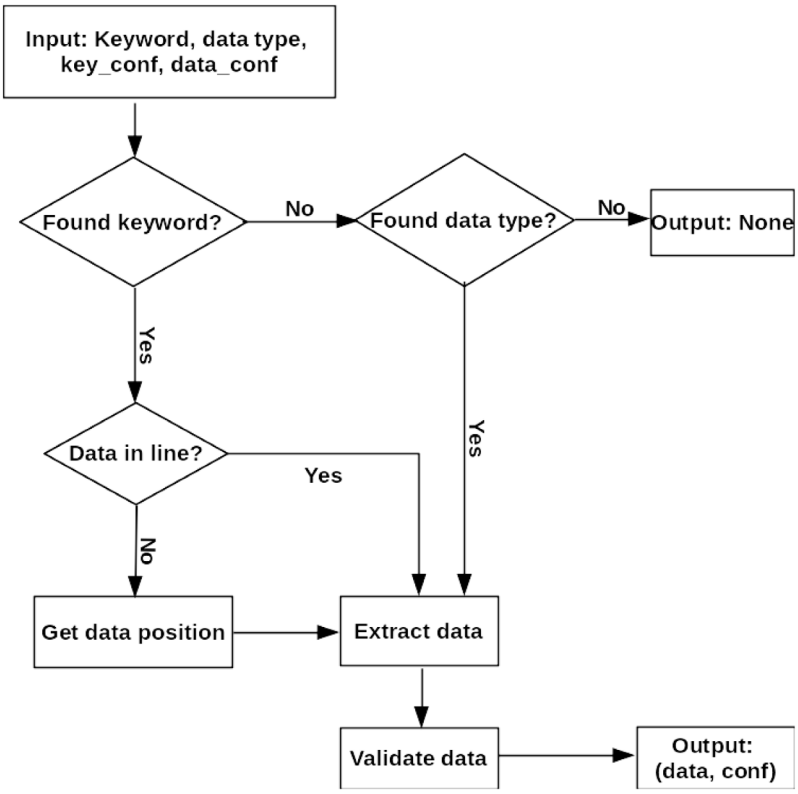


Fig. 4. The information extraction workflow process.

```
If block is vertically aligned or horizontally
    aligned with SELLER block and
    has no different content in [COMPANY,
        ADDRESS, ID, VAT NUMBER]
    compared to SELLER,
then label is SELLER.
```

Fig. 5. A buyer/seller classification rule example.

data. For example, the Czech VAT number begins with "CZ" and not "C2" or the email symbol is "@" and not "©".

The blocks with buyer and seller information are not always accompanied by keywords and their information may even be spread into several blocks. For those blocks, the information is classified into buyer's or seller's using a set of logical rules. These rules consider the block position in the page (e.g. prevalently, the seller's contact information is located at the bottom of the page), and alignment and content comparison with the found seller/buyer blocks. Finally, global rules based on the observation that the seller block often comes before the buyer block in vertical or horizontal direction are applied to blocks unprocessed in previous steps. Currently, OCRMiner uses 15 classification rules and 5 global rules written in descending confidence order. If a block satisfies a rule, then the corresponding label is returned and the system ignores the remaining rules. A classification rule example can be found in Fig. 5.

Different from other types of information which are mostly expressed in a text line, addresses usually spread in more than one line, in particular cases even with other text in between. Therefore, the address extraction does not simply extract an annotated *location* entity. In this step, the Libpostal address parser is employed with detailed address parts recognition. The extraction step takes "road" (street name) and "house_number" as important signpost to start an address. The labels are sorted in the order of "country", "city", "city_district", "suburb", "postcode", "road", "house_number", "house". The labels at the beginning of the list have higher rank than the labels at the end. Then, the text line containing the highest ranked label is considered as the end of the address. If the extracted address contains a label at the beginning of the list, i.e. a country or a city then the confidence number is higher than for the other labels (the address may miss some part of information). A LOC entity annotation also increases the confidence number. Example of an address block analysis is presented in Fig. 6.

4. Experiments and evaluation

4.1. The invoice dataset

The experimental invoice dataset is collected in cooperation with a renowned copy machine producer. Due to the sensitiveness of the invoice content, the dataset is not publicly available. The collection consists of two main parts: English invoices and Czech invoices, see Fig. 7. The English group consists of invoices from more than 50 suppliers all over the world whereas the Czech data comes from over 100 vendors in the Czech republic.

In each dataset, for developing and testing purposes 60 invoices were randomly selected. Those 120 invoices are from 76 vendors.

a)



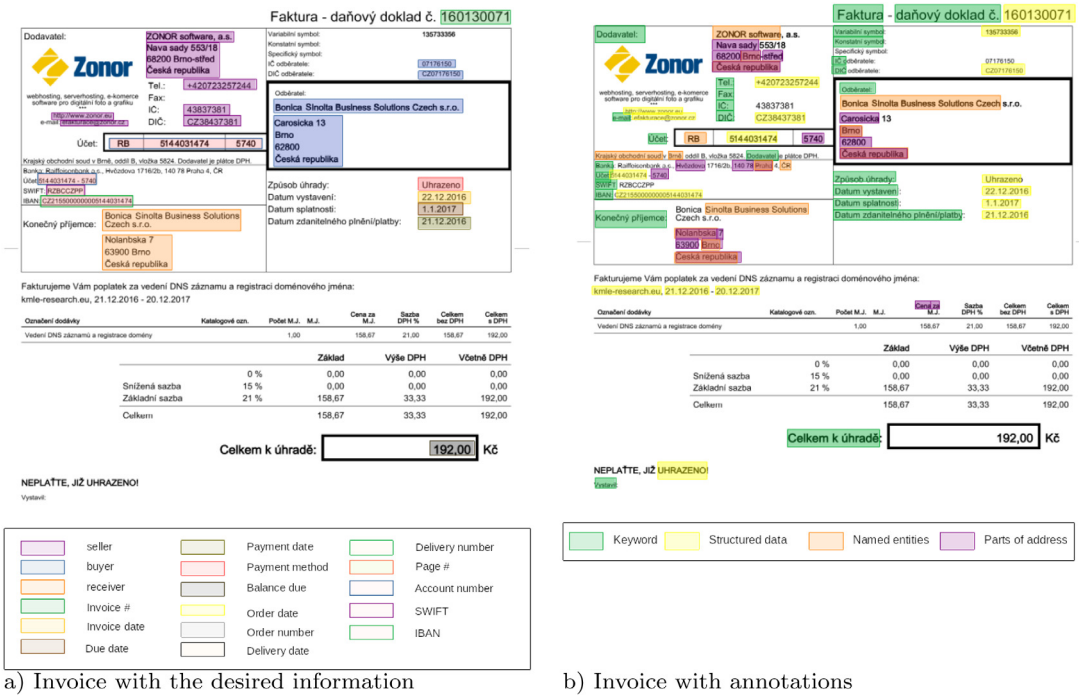
c)

```
<item field="BUYER" conf="0.9">
  <subitem field="COMPANY" value="Konica Minolta" conf="0.6"/>
  <subitem field="CONTACTS" value="Konica Minolta" conf="0.8"/>
  <subitem field="CONTACTS" value="Matej Dusik" conf="0.8"/>
  <subitem field="ADDRESS" value="Zaroskicka 13, 63900 Brno, Czech Republic" conf="0.9"/>
</item>
```

b)

```
<block id="2" bbox="310 727 352 770" left_blocks="" right_down_blocks="" bottom_blocks="4" right_blocks="3" top_blocks="1" abspos_y="top" abspos_x="left" type="buyer info">
  <line id="2" bbox="310 727 352 770"><annot type="buyer" src="detect_keyw" data_type="keyword"><annot type="state" src="detect_addr" data_type="data">To</annot></annot></line>
</block>
<block id="3" bbox="653 731 928 1002" left_blocks="2" right_down_blocks="" bottom_blocks="5" right_blocks="" top_blocks="" abspos_y="top" abspos_x="left" type="buyer info">
  <line id="3" bbox="655 731 872 758"><annot type="PERSON" src="detect_ner" data_type="data" language="en">Konica Minolta</annot></line>
  <line id="4" bbox="653 791 928 825"><annot type="road" src="detect_addr" data_type="data">Attn. <annot type="PERSON" src="detect_ner" data_type="data" language="en">Matej Dusik</
    annot></annot></line>
  <line id="5" bbox="653 850 878 878"><annot type="road" src="detect_addr" data_type="data"><annot type="PERSON" src="detect_ner" data_type="data" language="en">Zaroskicka</annot></annot>
    13</line>
  <line id="6" bbox="654 909 844 937"><annot type="postcode" src="detect_addr" data_type="data">63900</annot> <annot type="city" src="detect_addr" data_type="data"><annot type="PERSON"
    src="detect_ner" data_type="data" language="en">Brno</annot></annot></line>
  <line id="7" bbox="654 968 906 1002"><annot type="country" src="detect_addr" data_type="data"><annot type="LOCATION" src="detect_ner" data_type="data" language="en">Czech Republic</
    annot></annot></line>
</block>
```

Fig. 6. An example of an address block analysis: (a) the input image, (b) after annotation, and (c) the extracted information.



a) Invoice with the desired information

b) Invoice with annotations

Fig. 7. The input invoice dataset examples.

Each vendor has a different layout, even the layouts of invoices from the same vendor change over time. Ten out of the 60 are used as a development set. For instance, the ten English development invoices are by nine different suppliers in Austria, Poland, US, UK, the Netherlands, Germany, and Italy. More than a half of the 50 testing invoices are from 18 other suppliers which have not appeared in the development set. The annotated XML files expose detailed positional information of 7602 text blocks, 18,672 lines, and 9096 automated annotations of 64 annotation types. The files have also been manually annotated for the metadata items containing altogether 1523 values of 24 types for the information extraction process. A detailed overview of the information items is displayed in Table 2.

4.2. Evaluation

In the system evaluation, [10] the extracted field values are classified into 3 result classes. Each piece of information is evaluated as a *match*, a *partial match*, or a *mismatch*. For other data than company name and address, the *match* means always an exact match. With company name and address, the *match* classification allows to ignore differences in unimportant pieces of the gold standard values. A *match* is assigned if the extracted data contains the main information, even though some words are missing or surplus. For example, a company name of:

ground truth:

`Konica Minolta Business Solution Czech spol, s.r.o.`

extracted:

`Konica Minolta Business Solution Czech spol, s.r.0`

are considered a *match*. In this example, the letter "o" in "s.r.o." was recognized as "0" (zero) by the OCR engine. In this context, such OCR errors are easy to identify and fix as the "tail" of the name contains the known organization type ("s.r.o." in Czech corresponds to "Ltd." in English). Or the address:

[10] The information extraction task is evaluated independently from the page classification module which is fully described and evaluated in Section 3.2.

Table 2
The most common item types in the development and test sets.

Type of info	En dev set	En test set	Cz dev set	Cz test set
Invoice number	10	45	10	49
Order number	6	12	6	8
Swift	6	8	6	20
Account number	8	9	8	45
Page number	4	7	4	16
Invoice date	10	47	10	46
Due date	7	23	8	45
Payment date	8	10	8	37
Total due	9	27	9	25
IBAN	7	9	7	22
Payment method	8	18	8	34
Company name	23	101	23	114
Contacts	6	24	6	21
Address	25	101	25	119
Vat number	18	59	18	91
Email	8	24	8	35
Website	3	25	3	23
Phone number	15	17	15	40
Company id	0	0	20	99
Amount paid	0	21	0	0

Table 3
Information extraction evaluation results for the English set.

	regular expressions		similarity search	
	match	in %	match	in %
Match	523	84.22	527	**84.86**
Partial match	35	5.64	35	**5.64**
Mismatch	63	10.14	59	**9.50**
Total	621	100.00	621	100.00

ground truth:

```
Level 6, 341 George St, Sydney NSW 2000, Australia
```

extracted:

```
Atlassian Pty Ltd, Level 6, 341 George St, Sydney NSW 2000, Australia
```

are also regarded as *match*ing.

A *partial match* is assigned if the extracted information contains correct information, but some part is missing or some character is incorrectly recognized by the OCR tool. For instance, if the OCR tool recognized "0" character as "O" in the order number "BXOEF24CA4E2" instead of "BX0EF24CA4E2", or an address misses the street name:

ground truth:

```
Zarosicka 4395/13, Brno, Jihomoravsky kraj 62800, Czech Republic
```

extracted:

```
Brno, Jihomoravsky kraj 62800, Czech Republic
```

The *partial match* is "correctable" in the sense of finding regular OCR character mismatches or via the possibility to search for the expanded value in the identified block.

The comparison between the ground truth and the extracted information in the evaluation is first preprocessed automatically using the Levenshtein distance. The result returns a *match* if the distance is zero, a *partial match* if the edit distance is less than a threshold (set to 2 characters in the experiment) and a *mismatch* otherwise. Then, due to the complexity of address and company name blocks, the evaluation is double-checked manually. For the group-type information such as seller and buyer, the result counts are counted for each subfield, i.e. the company name, the address, and so on. This means that even if all information including company name, address, company id, vat number are correctly extracted but the group is wrongly classified as seller or buyer, then the evaluation module will count all the subfields as mismatches.

Table 4
Information extraction evaluation results with the Czech set.

	regular expressions		similarity search	
	match	in %	match	in %
Match	679	75.28	756	**83.81**
Partial match	30	3.33	40	**4.43**
Mismatch	193	21.40	106	**11.75**
Total	902	100.00	902	100.00

As can be seen in the evaluation of the page classification module, the keyword features are the most important in the correct identification of the block types. To be able to cope with OCR character misclassifications, OCRMiner contains two different modules to search for keywords. The first module uses regular expressions patterns and the second module is based on an approximate similarity search. Within the similarity search, the text substrings are evaluated with weighted edit distance (Levenshtein distance) measure which includes a small cost (0.1 in the experiments) for a substitution of common errors (such as 'l', 't' and 'f', or 'u' and 'v') and insertion or deletion of punctuation or white space, while any other edits have the cost of 1.0.[11] The substring of the text line which has the weighted edit distance less than a threshold (15% of the length of the keyword) is marked as the keyword. The results of these two modules for the English and Czech evaluation sets are presented in Tables 3 and 4. The similarity search module adapts better to OCR errors, which leads to slight improvements in both the languages when compared to the method using regular expressions match −0.64% or 3.22% of mismatches are solved or partially solved in the English and Czech datasets respectively. In general, the whole system works better for the English set where above 90% of ground truth items are correctly identified (with exact or partial match) compared to about 88% of the Czech invoice items.

A detailed analysis of the extraction accuracy for each field is illustrated in Fig. 8. Both datasets have high match ratios at the page number, the invoice date, the due date, and the payment method. The extraction works better for the English invoices with items such as the invoice number, the VAT number, IBAN, the total due amount, or the website. In contrast, the Czech dataset has better extraction results with the order number, the account number, the payment date, and the contact information. There is an exceptional low match ratio for the payment date in the English dataset due to OCR errors in a subset of English invoices from a particular provider. English invoices also witness a high partial match ratio with the order number. Different from Czech invoices, where the order number is a digital series, in the English dataset, the order number is often a mixture of numbers and letters, for example "AP0CDF89CC10". The first number "0" is between letters, and the OCR engine often misrecognizes it as the letter "O", causing the high partial match. Another high mismatch is the contact person item which is caused by the situation when NER misrecognized a person as an organization due to capitalized letters. A detailed error analysis is given in Section 4.3.

For a performance comparison, the OCRMiner dataset has been evaluated with the state-of-the-art InvoiceNet system[12] which is based on the work of Palm et al. [20]. InvoiceNet solves the end-to-end information extraction task using deep neural network models. Six common invoice fields were selected for the evaluation: the invoice number, the invoice date, the seller's name and address and the buyer's name and address. Each field requires a separate model. Since the original pre-trained models are not publicly available, InvoiceNet has

[11] See [44] for more details.
[12] https://github.com/naiveHobo/InvoiceNet.

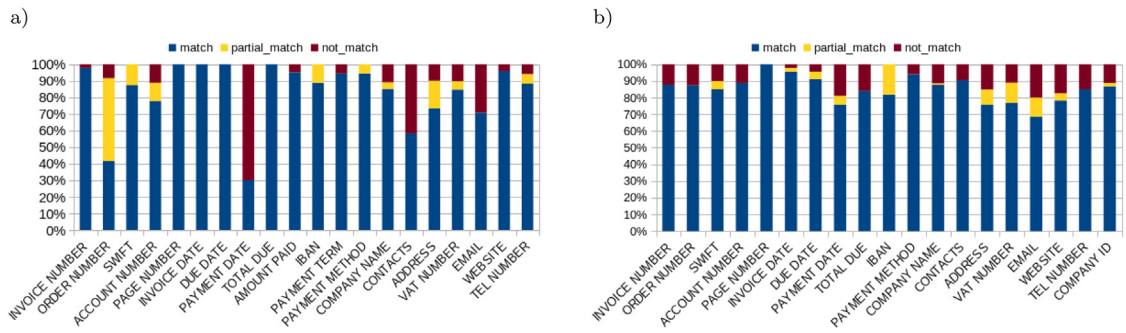


Fig. 8. Evaluation of individual fields for (a) the English invoices, and (b) the Czech invoices.

Table 5
Performance comparison on specific fields in %.

		English data			Czech data	
field	measure	InvNet1	InvNet2	OCRMiner	InvNet3	OCRMiner
inv_number	match	–	46.67	**97.78**	83.67	**87.76**
	partial match	–	0.00	0.00	4.08	0.00
	mismatch	–	53.33	2.22	12.24	12.24
inv_date	match	0.00	0.00	**100.0**	0.00	**95.65**
	partial match	0.00	0.00	0.00	0.00	2.17
	mismatch	100.0	100.0	0.00	100.0	2.17
seller_name	match	12.24	44.90	**83.67**	77.27	**79.55**
	partial match	0.0	2.04	0.00	9.09	9.09
	mismatch	87.76	53.06	16.33	13.46	11.36
seller_addr	match	4.08	6.25	**81.63**	2.00	**86.00**
	partial match	16.33	87.50	14.29	82.00	4.00
	mismatch	79.59	6.25	4.08	16.00	10.00
buyer_name	match	–	71.11	**86.67**	82.00	**84.00**
	partial match	–	2.22	8.89	8.00	2.00
	mismatch	–	26.67	4.44	10.00	14.00
buyer_addr	match	–	2.22	**66.67**	0.00	**80.00**
	partial match	–	**93.33**	20.00	30.00	8.00
	mismatch	–	4.44	13.33	70.00	12.00

Table 6
Evaluation of location detection by the combination of Libpostal and Named entity recognition (NER) modules.

English set:	Libpostal		NER	
both modules	287	25%	287	83%
one module only	876	75%	57	17%
– true positives	230	20%	23	7%
– false positives	646	55%	34	10%

Czech set:	Libpostal		NER	
both modules	165	12%	165	59%
one module only	1223	88%	117	41%
– true positives	292	21%	116	41%
– false positives	931	67%	1	0%

Table 7
Ablation study with the English dataset.

Features	Match	Partial match	Mismatch
Full model	84.86%	5.64%	9.50%
Without keywords	61.06%	3.04%	35.90%
Without data	49.20%	1.12%	49.68%

now been trained in three possible setups. First, the InvoiceNet models have been trained with the SROIE datasets [3] including 727 receipts (denoted as InvNet1). The second system (InvNet2) has been trained with 208 English invoices of the OCRMiner dataset and the last one has been trained with 500 Czech invoices from the same dataset (InvNet3). InvNet1 and InvNet2 are evaluated with the English test set whereas InvNet3 is evaluated with the Czech test set. The results are summarized in Table 5.

Overall, OCRMiner performs better than InvoiceNet in all fields with both the English and the Czech dataset. InvoiceNet locates the position of addresses in English invoices better than OCRMiner but fails to extract the full data. In multiple lines, multi-word text data (e.g. an address or a company name) are identified only partially with important parts missing. In most cases, only the street name and house number are extracted. The results also show that a system trained with receipts cannot be generalized to invoices.

Table 6 displays an evaluation of the location detection by the two modules employed in the process — Libpostal and the named entity recognition (NER) module. Both modules mostly agree on city and country names. Libpostal offers detailed annotation types of "road", "postcode", "suburb", "city_district", "city", "state district", "state", and "country" labels. Of these, the postcode and street are the most important regarding the final scoring. Consequently, Libpostal module has a higher recall than the NER model, while the NER module offers better precision. Libpostal is also not so sensitive to uppercasing, e.g. BRNO (a city), or CZECH REPUBLIC. This is why an ensemble module is used in the system which allows to take advantage of each model's capabilities. The system also requires a combined information support of at least two detected address items in the analysed text block to eliminate most of false positive errors.

To better understand the contribution of keyword and data annotations to the extraction, a set of ablation test were run for OCRMiner. In each test, one of those two annotation types is left out and the information extraction is based on the remaining annotations. The extraction process can still identify some fields but with a decreased value of the confidence score. The overall results evaluated with the English test set are shown in Table 7. As expected, when keywords are missing, the match ratio has fallen off for fields relying on the keyword only, such as the invoice number, the SWIFT code, or the account number with the drop of 97.78%, 87.75%, and 77.78% respectively. The textual fields such as the company name or address and the contact person are slightly affected with the drop of 0.99%, 4.16%, and 6.93% respectively. In contrast, these fields' *match* ratios drop quickly (by 62.38%, 58.33%, and 73.27%) when the data annotations are absent.

4.3. Error analysis

A detailed analysis of errors in the system output was evaluated based on the results of the similarity search module. The causes of errors can be categorized into five groups, but their ratios in each dataset are not the same.

Different from other pieces of extracted information, the seller/buyer/delivery addresses are groups of information. They are often not denoted by specific keywords, fragmented into several blocks and their exact distinction is left to the reader. To extract this information, the system needs to classify the specified blocks into the seller, the

Table 8
Detailed analysis of the error types.

Mismatches				
Error types	En set		Cz set	
Misclassified role	27%	16	**33%**	35
OCR errors	10%	6	10%	11
Misclassified NE	**41%**	24	7%	7
New keywords	12%	7	12%	13
Layout	3%	2	17%	18
Other	7%	4	21%	22
Total	100%	59	100%	106

Partial matches				
Error types	En set		Cz set	
OCR errors	34%	12	**66%**	29
Addresses	**46%**	16	14%	6
Other	20%	7	20%	9
Total	100%	35	100%	44

buyer and the delivery address based on their position and content. 27% of errors in the English invoices and 35% of errors in the Czech ones are correctly extracted but misclassified in these three categories. The combination of reasons for this type of errors includes errors in detecting the addresses and named entities (false positives), insufficient rule details, and an extra variety in the layout formats (e.g. the position relation between the seller and the buyer without a *keyword* distinction) that makes the global rules conflicting in the difficult cases.

While the misclassification causes the largest portion of errors in the Czech invoices (33%), incorrect identification of entities forms the main reason of mismatches in the English invoices (41% of errors) including persons classified as organizations ("PETR GOTTHARD", "EVA LYSONKA", ...) or undetected organizations and locations.

OCR errors and unknown keywords cause almost the same portion of errors in each dataset with 10% and 12% respectively. The Czech invoices display a broad range of layout formats which is projected to 17% of layout errors compared to only 3% in the English set. Most of these errors happen with special cases when the keyword and the data are in an unexpected relative position, e.g. the keyword below the data, or one line of a company info in a distant stand-alone block.

The error analysis of partial matches revealed that the OCR errors cause 64% of errors for English, and 34% for the Czech invoice items. A significant portion of these errors happens in email addresses where the "at" symbol "@" is not recognized correctly in the Czech invoices with the output being recognized as "&", "&&", "Q", or "©". Another typical OCR error is a duplicated number. For example, in the VAT number "CZ00176150" the "0" is doubled to "CZ000176150". Besides OCR errors, a missing part of address is the reason for 46% of partial matches within the English invoices and for 16% in the Czech invoices respectively. The extraction module only looks for (parts of) addresses in continuous lines. Therefore, in cases where some other pieces of information are inserted in those lines, e.g. a VAT number, a part of the address is misidentified. An overview of all the error types is presented in Table 8.

5. Conclusions and future directions

In this paper, we have presented the current results of the OCRMiner system for extracting information from semi-structured scanned business documents. OCRMiner is designed to process the document in the same way as a human reader employing a combination of text analysis techniques and positional layout features.

The system has been evaluated with two invoice datasets (English and Czech) containing data from more than 150 different vendors from all over the world. Using an open source OCR, with a small development set consisting of only 20 invoices, the system is able to detect **90.5%** items in the English invoices and **88.2%** in the Czech invoices with 93%–96% of these item values being correct matches. OCRMiner is also able to detect whether an image is (the first page of) an invoice or not with **98%** accuracy.

Such result is fully comparable to the state-of-the-art systems which are, however, trained on much larger datasets. The main advantages of OCRMiner lie in employing rich text analysis annotations combined with the positional properties of each text block. With underlying language dependent modules (OCR, keywords and named entities), the higher-level modules are language independent and offer a straightforward application to a new language without a need of huge annotated invoice datasets. The system is also able to cope with OCR errors using similarity search keyword detection.

The system tries to mimic the extraction procedure used by humans when reading an invoice. Even when the layout changes, the principle that the data is located around keywords (often on the right or under, or in some cases, above) should not change. This allows OCRMiner to adapt to most possible new layout varieties. Although, as is mentioned in the error analysis, the rules have limited effect in distinguishing the buyer and seller blocks when there are no keywords attached to them. In this case, the rules cover the most frequent possibilities, but with some invoices they may even conflict with the designer's decision. For example, the seller mostly appears on the left of the buyer but, in some rare cases, the seller and buyer position are swapped. In the future version, an ensemble seller/buyer distinguishing model with a supervised learning method will be tested as preliminary experiments show promising results.

Another possibility to improve the performance lies in the layout analysis step. A portion of errors are caused by block fragmentations where the desired data is disconnected from the keywords and the surrounding context. Identification of such remote pieces of information as e.g. parts of the seller or buyer address can further boost the accuracy of extracted information. One possible direction is taking the advantage of drawn tabular lines that separate invoice parts but that are ignored by the OCR process.

In the ongoing research, OCRMiner is currently adapted to the information extraction task from other types of business documents such as contracts, reports, orders, or meeting minutes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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